

Appendix

eMethods

eTable.1. Missing Value Statistics of the 40 preliminarily selected predictors

eTable.2. Codes of candidate predictors in the NACC

eTable.3. The baseline characteristics of the NACC subjects included in the study

eTable.4. The result of hierarchical clustering dendrogram of predictors

eTable.5. AUC values for different years of each neuropsychological measurement scale

eTable.6. Overall evaluation criteria for each model-12 predictors (excluding ApoE ϵ 4)

eTable.7. AUC values for different years of each model-12 predictors (excluding ApoE ϵ 4)

eFig.1. Pairwise relationship diagrams of each predictor in the dataset

eFig.2. Correlations heatmap of 13 predictors

eMethods

Description of primary cognitive tests

NACCMOCA: The MoCA includes items to assess memory, language, visuospatial ability, attention, and executive function, and the total score is 30 points.

CDRGLOB: Scores on the memory, judgment and social activities dimensions of CDR scale were included as predictors.

CRAFTURS, CRAFTDVR: The Craft Story 21 Recall – Immediate was used to assess verbal episodic memory ability. After the examiner reads a story to the subject, the subjects are asked to retell the story from memory. In Craft Story 21 Recall – Delayed, subjects are asked to repeat the story after a delay of 20 minutes.

ANIMALS, VEG: The Category Fluency score is a widely used measure of semantic memory. Subjects are asked to name a given semantic category (animals or vegetables) and score the named quantities.

UDSBENTD: The Benson Complex Figure Copy assesses the visual constructional ability by asking subjects to reproduce a complex geometric figure and then recall the design 10 to 15 min later.

TRAILA, TRAILB: The Trail Making Test is an accessible neuropsychological instrument that provides the examiner with information on a wide range of cognitive skills and can be completed in 5–10 min[1]. Trail Making Test A: attention, Trail Making Test B: executive function.

MINTTOTS: Multilingual Naming Test[2]. The MINT consists of 32 black-and-white line drawings of objects that are presented one at a time to be named beginning with relatively easier items and ending with relatively more difficult to name items. If an individual encounters difficulty identifying an object, a semantic or phonemic cue is provided. The number of spontaneous correct responses and correct responses following a semantic cue are summed to give a total score. Thus, the abbreviated MINT total scores range from 0 to 32 and these were used as the primary outcome measure. The Boston Naming Test is similar in function to the Multilingual Naming Test.

- [1] Bowie CR, Harvey PD. Administration and interpretation of the trail making test. Nat Protoc.2006;1:2277–81. <https://doi.org/10.1038/nprot.2006.390>.
 [2] Stasenko A, Jacobs DM, Salmon DP, Gollan TH. The multilingual naming test (MINT) as a measure of picture naming ability in alzheimer’s disease. J Int Neuropsychol. Soc : JINS 2019;25:821–33. <https://doi.org/10.1017/S1355617719000560>.

eTable.1. Missing value statistics of the 40 preliminarily selected predictors

Predictors	The number of valid values	The number of missing values
NACCAGE	3789	0
NACCGDS	3602	187
NACCBMI	3423	366
HRATE	3317	472
BPSYS	3289	500
EDUC	3769	20
ANIMALS	3646	143
VEG	3625	164
TRAILA	3587	202

NACCMOCA	3220	569
MOCATOTS	3237	552
TRAILB	3432	357
CRAFTURS	3257	532
CRAFTDVR	3249	540
MINTTOTS	3232	557
UDSBENTD	3475	314
CDRGLOB	3789	0
SEX	3789	0
RACE	3769	20
MARISTAT	3775	14
INDEPEND	3764	25
COGMODE	3617	172
COGSTAT	3692	97
DECIN	3636	153
JUDGMENT	3789	0
NACCCOGF	3576	213
MOMODE	3771	18
DECSUB	3779	10
MEMORY	3789	0
COGJUDG	3772	17
DEP	3789	0
COGVIS	3751	38
NACCMOTF	3778	11
BEMODE	3715	74
NACCBEHF	3759	30
COMMUN	3789	0
COGMEM	3782	7
NACCNE4S	3237	552
RESIDENC	3769	20
COURSE	3354	435

The above are the names of 40 preliminarily predictors.

The codes of the 40 preliminarily selected predictors are shown in Supplementary **eTable.2**.

eTable.2. Codes of candidate predictors in the NACC

Predictors	Short descriptor	Label
SEX	Subject's sex	1=Male, 2=Female
RACE	Race	1=White 2=black or African American 3=American Indian or Alaska Native 4=Native Hawaiian or Other Pacific Islander 5=Asian 50=Other (specify) 99=Unknown
EDUC	Years of education	0 –36, 99=Unknown
MARISTAT	Marital status	1=Married 2=Widowed (丧偶的) 3=Divorced (离婚的) 4=Separated (分离的) 5=Never married (or marriage was annulled) 6=Living as married/domestic partner 9=Other or unknown
INDEPEND	Level of independence	1 = Able to live independently 2 = Requires some assistance with complex activities 3 = Requires some assistance with basic activities 4 = Completely dependent 9 = Unknown
NACCNIHR	Derived NIH race definitions	1 = White, 2 = black or African American, 3 = American Indian or Alaska Native, 4 = Native Hawaiian or Pacific Islander, 5 = Asian, 6 = Multiracial, 99 = Unknown or ambiguous
TOBAC30	Smoked cigarettes in last 30 days	0=No 1=Yes 9=Unknown -4= Not available

SMOKYRS	Total years smoked cigarettes	0 – 87 88=Not applicable 99=Unknown -4= Not available
DIABETES	Diabetes	0 = Absent, 1 = Recent/Active, 2 = Remote/Inactive ,9 = Unknown, 4 = Not available
HYPERTEN	Hypertension	0 = Absent, 1 = Recent/Active, 2 = Remote/Inactive ,9 = Unknown, 4 = Not available
HYPERCHO	Hypercholesterolemia	0 = Absent, 1 = Recent/Active, 2 = Remote/Inactive ,9 = Unknown, 4 = Not available
B12DEF (	Vitamin b12 deficiency	0 = Absent, 1 = Recent/Active, 2 = Remote/Inactive ,9 = Unknown, 4 = Not available
THYROID	Thyroid disease	0 = Absent, 1 = Recent/Active, 2 = Remote/Inactive ,9 = Unknown, 4 = Not available
COGSTAT	Per clinician, based on the neuropsychological examination, the subject’s cognitive status is deemed	0 = Clinician unable to render opinion 1 = better than normal for age 2 = Normal for age 3 = One or two test scores abnormal 4 = Three or more scores are abnormal or lower than expected 9 = Missing-4 = Not available
COGMODE	Mode of onset of cognitive symptoms Numeric longitudinal	0 = No impairment in cognition 1 = Gradual 2 = Subacute 3 = Abrupt 4 = Other (specify) 99 = Unknown
HIPPATR	Hippocampal atrophy	0 = No 1 = Yes 8 = Unknown
TRAILA	Trail Making Test Part A — Total number of seconds to complete	0 – 150 995 = Physical problem 996 = Cognitive/behavior problem 997 = Other problem 998 = Verbal refusal 4 = Not available
TRAILB	Trail Making Test Part B — Total number of seconds to complete	0 – 300 995 = Physical problem 996 = Cognitive/behavior problem 997 = Other problem 998 = Verbal refusal 4 = Not available

DECIN	Does the co-participant report a decline in subject's memory (relative to previously attained abilities)?	0 = No
		1 = Yes
		8 = There is no co-participant
		9 = Unknown
JUDGMENT	Judgment and problem-solving	0.0 = No impairment
		0.5 = Questionable impairment
		1.0 = Mild impairment
		2.0 = Moderate impairment
		3.0 = Severe impairment 无
NACCCOGF	Indicate the predominant symptom that was first recognized as a decline in the subject's cognition	0 = No impairment in cognition
		1 = Memory
		2 = Orientation
		3 = Executive function — judgment, planning, problem-solving
		4 = Language
		5 = Visuospatial function
		6 = Attention/concentration
		7 = Fluctuating cognition
		8 = Other (specify)
NORMEXAM	Were there abnormal neurological exam findings?	99 = Unknown
		0 = Abnormal findings
		1 = No abnormal findings or findings normal for age
		9 = Unknown
CRAFTURS	Craft Story 21 Recall (Immediate) — Total story units recalled, paraphrase scoring	-4 = Not available
		0 – 25
		95 = Physical problem
		96 = Cognitive/behavior problem
		97 = Other problem
		98 = Verbal refusal
VEG	Vegetables — Total number of vegetables named in 60 seconds	-4 = Not available
		0-77
		95 = Physical problem
		96 = Cognitive/behavior problem
		97 = Other problem
ANIMALS	Animals — Total number of animals named in 60 seconds	98 = Verbal refusal-4 = Not available
		0-77
		95 = Physical problem
		96 = Cognitive/behavior problem

		97 = Other problem
		98 = Verbal refusal-4 = Not available
MINTTOTS	Multilingual Naming Test (MINT)	0-32
	— Total score	95 = Physical problem
		96 = Cognitive/behavior problem
		97 = Other problem
		98 = Verbal refusal
		-4 = Not available:
UDSBENTD	Total score for 10- to 15-minute delayed drawing of benson figure	0-17
		95 = Physical problem
		96 = Cognitive/behavior problem
		97 = Other problem
		98 = Verbal refusal
		-4 = Not available:
DECCLCOG	Based on the clinician's judgment, is the subject currently experiencing meaningful impairment in cognition?	0 = No
		1 = Yes
		-4 = Not available
MOMODE	Mode of onset of motor symptoms	0 = No motor symptoms
		1 = Gradual
		2 = Subacute
		3 = Abrupt
		4 = Other
		99 = Unknown
CRAFTDTI	Craft Story 21 Recall (Delayed) — Delay time	0-85
		99 = Unknown
		-4 = Not available
DECSUB	Does the subject report a decline in memory (relative to previously attained abilities)?	0 = No
		1 = Yes
		8 = Could not be assessed/subject too impaired
		9= Unknown
OTHCOG	Presumptive etiologic diagnosis — Other neurological, genetic, or infectious condition	0 = No
		1 = Yes
		-4 = Not applicable
MEMORY	Memory	0.0 = No impairment
		0.5 = Questionable impairment
		1.0 = Mild impairment

2.0 = Moderate impairment
3.0 = Severe impairment

COGORI	Indicate whether the subject currently is meaningfully impaired, relative to previously attained abilities, in orientation	0 = No 1 = Yes 9 = Unknown -4 = Not available
COGJUDG	Indicate whether the subject currently is meaningfully impaired, relative to previously attained abilities, in executive function — judgment, planning, or problem-solving	0 = No 1 = Yes 9 = Unknown
DEP	Presumptive etiologic diagnosis of the cognitive disorder — Depression	0 = No 1 = Yes
COURSE	Overall course of decline of cognitive/ behavioral/motor syndrome	1 = Gradually progressive 2 = Stepwise 3 = Static 4 = Fluctuating 5 = Improved 8 = Not applicable 9 = Unknown
MEDSIF	Primary, contributing, or non-contributing cause of cognitive impairment — medications	1 = Primary 2 = Contributing 3 = Non-contributing 7 = Cognitively impaired but no diagnosis of impairment due to medications 8 = Diagnosis of normal cognition -4 = Not applicable: UDS form submitted did not collect data in this way, or a skip pattern precludes response to this question
COGVIS	Indicate whether the subject currently is meaningfully impaired, relative to previously attained abilities, in visuospatial function	0 = No 1 = Yes 9 = Unknown

CDRLANG	Language	0.0 = No impairment 0.5 = Questionable impairment 1.0 = Mild impairment 2.0 = Moderate impairment 3.0 = Severe impairment -4 = Not available:
NACCMOTF	Indicate the predominant symptom that was first recognized as a decline in the subject's motor function	0 = No motor symptoms 1 = Gait disorder 2 = Falls 3 = Tremor 4 = Slowness 99 = Unknown
BEMODE	Mode of onset of behavioral symptoms Numeric longitudinal	0 = No behavioral symptoms 1 = Gradual 2 = Subacute 3 = Abrupt 4 = Other (specify) 99 = Unknown
MOCATOTS	MoCA Total Raw Score — uncorrected	0 – 30 88 = Item(s) or whole test not administered -4 = Not available
MOCAREGI	MoCA: Memory — Registration (two trials)	0-10 95 = Physical problem 96 = Cognitive/behavior problem 97 = Other problem 98 = Verbal refusal -4 = Not available
DECCLMOT	Based on the clinician's judgment, is the subject currently experiencing any motor symptoms?	0 = No 1 = Yes -4 = Not available
DIGBACCT	Number Span Test: backward — Number of correct trials	0-14 95 = Physical problem 96 = Cognitive/behavior problem 97 = Other problem 98 = Verbal refusal -4 = Not available

NACCBEHF	Indicate the predominant symptom that was first recognized as a decline in the subject's behavior	0 = No behavioral symptoms 1 = Apathy/withdrawal 2 = Depressed mood 3 = Psychosis 4 = Disinhibition 5 = Irritability 6 = Agitation 7 = Personality change 8 = REM sleep behavior disorder 9 = Anxiety 10 = Other (specify) 99 = Unknown
COMMUN	Community affairs	0.0 = No impairment 0.5 = Questionable impairment 1.0 = Mild impairment 2.0 = Moderate impairment 3.0 = Severe impairment
CORTIF	Primary, contributing, or non-contributing cause of cognitive impairment — Corticobasal degeneration (CbD)	1 = Primary 2 = Contributing 3 = Non-contributing 7 = Cognitively impaired but no CbD diagnosis 8 = Diagnosis of normal cognition
COGMEM	Indicate whether the subject currently is meaningfully impaired, relative to previously attained abilities, in memory	0 = No 1 = Yes 9 = Unknown
FDGAD	FDG-PET pattern of AD	0 = No 1 = Yes 8 = Unknown/not assessed -4 = Not applicable
AMYLPET	Abnormally elevated amyloid on PET	0 = No 1 = Yes 8 = Unknown/not assessed -4 = Not applicable
AMYLCSF	Abnormally low amyloid in CSF	0 = No 1 = Yes 8 = Unknown/not assessed

-4 = Not applicable

NACCNE4S	Number of APOE e4 alleles	0 = No e4 allele 1 = 1 copy of e4 allele 2 = 2 copies of e4 allele 9 = Missing/ unknown/ not assessed
HYPOSOM	Hypsomnia/insomnia present	0 = No 1 = Yes 8 = Not assessed -4 = Not available
NACCBMI	Body mass index (BMI)	10.0 – 100.0 888.8 = Unknown or not assessed - 4 = Not available
NACCM MSE	Total MMSE score (using D-L-R-O-W)	0 – 30 88 = Score not calculated; missing at least one MMSE item 95 = Physical problem 96 = Cognitive/behavior problem 97 = Other problem 98 = Verbal refusal -4 = Not available:
BPSYS	Subject blood pressure (sitting), systolic	70 – 230 777 = blood pressure addendum submitted 888 = Unknown or not assessed - 4 = Not available
HRATE	Subject resting heart rate (pulse)	33 –160 888 = Unknown or not assessed - 4 = Not available
NACCGDS	Total GDS Score	0 – 15 88 = Could not be calculated -4 = Not available
NACCU DSD*	Cognitive status at UDS visit	1 = Normal cognition 2 = Impaired-not-MCI 3 = MCI 4 = Dementia
NACCALZD*	Presumptive etiologic diagnosis of the cognitive disorder— Alzheimer’s disease	0 = No 1 = Yes 8 = No cognitive impairment

Note: *The variables of survival events. The subject’s cognitive status is determined at every visit.

Please refer to the webpage of NACC for detailed information (<https://naccdata.org/>).

Abbreviations: AD = Alzheimer’s Disease

eTable.3. The baseline characteristics of the NACC subjects included in the study

Participants Characteristics	Overall (n=3674)	Non-AD group ^c (n=2710)	AD group ^d (n=964)	p-value	Univariate cox regression	
					Hazrad ratio	p-value
Follow up, month	24.5[13.4-40.4]	24.8[13.4-41.6]	23.7[13.3-37.5]	<0.001		
Age, year	75[69-81]	75[69-80]	76[70-82]	<0.001	1.020[1.013-1.028]	<0.001
Geriatric Depression Scale, score	2[1-3]	2[1-3]	2[1-3]	0.860	1.027[1.001-1.054]	0.043
BMI	26.4[23.7-29.5]	26.8[24.1-30]	25.4[23-28.18]	<0.001	0.937[0.923-0.950]	<0.001
Education, year	16[14-18]	16[14-18]	16[14-18]	0.115	1.009[0.988-1.030]	0.415
Category Fluency-ANIMALS, score	16[13-19]	17[13-20]	15[12-18]	<0.001	0.926[0.914-0.939]	<0.001
Category Fluency- Vegetables, score	11[8-13]	11[9-14]	10[7-12]	<0.001	0.900[0.885-0.917]	<0.001
Trail Making Test Part A, score	38[29-49]	37[29-48]	39[30-52]	<0.001	1.010[1.008-1.013]	<0.001
Trail Making Test Part B, score	112[80-176]	109[79-169]	120[85-192]	<0.001	1.003[1.002-1.004]	<0.001
Montreal Cognitive Assessment, score	23[21-25]	23[21-25]	22[19-24]	<0.001	0.876[0.861-0.891]	<0.001
Craft Story 21 Recall-Immediate, score	12[9-15]	13[10-15]	11[7-13]	<0.001	0.891[0.878-0.904]	<0.001
Craft Story 21 Recall-Delayed, score	11[7-16]	12[8-17]	7[2-12]	<0.001	0.897[0.888-0.907]	<0.001
Multilingual Naming Test, score	29[27-31]	29[27-31]	29[26-30]	<0.001	0.949[0.934-0.964]	<0.001
Benson Complex Figure Copy, score	8[5-10]	8[6-11]	6[2-8]	<0.001	0.847[0.834-0.861]	<0.001
SEX[Female]	1849(50.3%)	1364(50,3%)	485(50,3%)	0.991	1.004[0.885-1.139]	0.955
RACE[White] ^a	3020(82.2%)	2174(80.2%)	846(87.8%)	<0.001	2.087[1.652-2.637]	<0.001
Reported memory decline ^b	2854(77.7%)	1973(72.8%)	881(91.40%)		3.564[2.846-4.464]	<0.001
Clinical Dementia Rating-Global						
No impairment	388(10.6%)	357(13.2%)	31(3.2%)		reference	
Questionable impairment	3238(88.1%)	2336(86.2%)	902(93.6%)	<0.001	3.681[2.573-5.266]	<0.001
Mild impairment	48(1.3%)	17(0.6%)	31(3.2%)		18.440[11.195-30.372]	<0.001
Clinical Dementia Rating-JUDGMENT						
No impairment	1757(47,8%)	1415(52.2%)	342(35.5%)	<0.001	reference	
Questionable impairment	1672(45.5%)	1153(42.5%)	519(53.8%)		2.017[1.759-2.313]	<0.001

Mild impairment	242(6.6%)	141(5.2%)	101(10.5%)		4.076[3.257-5.101]	<0.001
Moderate impairment	3(0.1%)	1(0.0%)	2(0.2%)		64.543[15.832-263.132]	<0.001
Clinical Dementia Rating-MEMORY						
No impairment	476(13.0%)	440(16.2%)	36(3.7%)		reference	
Questionable impairment	2605(70.9%)	1973(72.8%)	632(65.6%)		3.167[2.263-4.431]	<0.001
Mild impairment	589(16.0%)	297(11.0%)	292(30.3%)	<0.001	8.724[6.170-12.337]	<0.001
Moderate impairment	4(0.1%)	0(0.0%)	4(0.4%)		108.122[38.190-306.112]	<0.001
Clinical Dementia Rating-COMMUN						
No impairment	2653(72.2%)	2086(77.0%)	567(58.8%)		reference	
Questionable impairment	934(25.4%)	581(21.4%)	353(36.6%)	<0.001	2.377[2.080-2.717]	<0.001
Mild impairment	84(2.3%)	42(1.5%)	42(4.4%)		4.640[3.384-6.362]	<0.001
Moderate impairment	3(0.1%)	1(0.0%)	2(0.2%)		4.762[1.187-19.100]	0.028
Number of APOE e4 alleles						
No e4 allele	2225(60.6%)	1788(66.0 %)	437(45.3%)		reference	
1 copy of e4 allele	1213(33.0%)	802(29.6%)	411(42.6%)	<0.001	1.742[1.523-1.994]	<0.001
2 copies of e4 allele	236(6.4%)	120(4.4%)	116 (12.0%)		2.404[1.959-2.950]	<0.001
Heart rate, frequency	67[60-72]	67[60-73]	66[60-72]	0.009	0.990[0.983-0.998]	0.009
Systolic blood pressure, mmHg	136[125-146]	136[125-146]	134[124-144]	0.015	0.995[0.991-0.999]	0.015
MoCA Total Raw Score, score	67[60-72]	67[60-73]	66[60-72]	0.009	0.990[0.983-0.998]	0.009
First predominant symptom						
Memory	2886(78.6%)	2044(75.4 %)	842(87.3%)		reference	
Orientation	26(0.7%)	22(0.8%)	4(0.4%)		0.441[0.152-1.285]	0.134
Executive function	344 (9.4%)	311(11.5%)	33 (3.4%)		0.258[0.178-0.372]	<0.001
Language	225 (6.1%)	165 (6.1%)	60 (6.2%)	<0.001	0.883[0.65-1.199]	0.425
Visuospatial function	64 (1.7%)	52 (1.9%)	12 (1.2%)		0.56[0.298-1.055]	0.073
Attention/concentration	113 (3.1%)	101 (3.7%)	12 (1.2%)		0.288[0.158-0.528]	<0.001
Fluctuating cognition	6 (0.2%)	0 (0%)	6 (0.2%)		0[0]	0.999
Other (specify)	10 (0.3%)	9 (0.3%)	1(0.1%)		0.27[0.034-2.132]	0.214
Mode of onset of motor symptoms						
No motor symptoms	2959(80.5%)	2163(79.8%)	796(82.6 %)		reference	
Gradual	654(17.8%)	494(18.2%)	160(16.6%)		0.88[0.724-1.071]	0.201
Subacute	14(0.4%)	11(0.4%)	3(0.3%)	0.098	0.741[0.206-2.663]	0.646
Abrupt	30(0.8%)	27(1.0%)	3(0.3%)		0.302[0.091-0.998]	0.05
Other	17(0.5%)	15(0.6%)	2(0.2%)		0.362[0.083-1.588]	0.178
Subject report a decline in memory	2928(79.7%)	2095(77.3%)	833(86.4%)	<0.001	1.867[1.521-2.291]	<0.001
Subject is impaired in executive function	1684(45.8%)	1171(43.2%)	513(53.2%)	<0.001	1.495[1.290-1.733]	<0.001

Diagnosis of cognitive disorder-Depression	693(18.9%)	530(19.6%)	163(16.9%)	0.071	0.837[0.690-1.016]	0.071
Subject is impaired in visuospatial function	551(15.0%)	1171(13.2%)	513(53.2%)	<0.001	1.495[1.290-1.733]	<0.001
Mode of onset of behavioral symptoms						
No behavioral symptoms	2099(57.1%)	1550(57.2%)	549(57.0%)		reference	
Gradual	1469(40.0%)	1069(39.4%)	400(41.5%)		1.056[0.909-1.228]	0.475
Subacute	58(1.6%)	48(1.8%)	10 (1.0%)	<0.046	0.588[0.296-1.171]	0.131
Abrupt	17(0.5%)	16(0.6%)	1(0.1%)		0.176[0.023-1.334]	0.093
Other (specify)	31(0.8%)	27(1.0%)	4(0.4%)		0.418[0.146-1.201]	0.105
NACCBHF						
No behavioral symptoms	2143(58.3%)	1581(58.3%)	562(58.3 %)		reference	
Apathy/withdrawal	158 (4.3%)	100(3.7%)	58(6.0%)		1.632[1.164-2.287]	0.004
Depressed mood	500(13.6%)	385(14.2%)	115(11.9%)		0.840[0.668-1.057]	0.137
Psychosis	14(0.4%)	12(0.4%)	2(0.2%)		0.469[0.105-2.101]	0.322
Disinhibition	30(0.8%)	24(0.9%)	6(0.6%)		0.703[0.286-1.729]	0.443
Irritability	315(8.6%)	218(8.0%)	97(10.1%)	<0.001	1.252[0.967-1.620]	0.088
Agitation	30(0.8%)	24(0.9%)	6(0.6%)		0.703[0.286-1.729]	0.443
Personality change	33(0.9%)	29(1.1%)	4(0.4%)		0.388[0.136-1.109]	0.077
REM sleep behavior disorder	100(2.7%)	90(3.3%)	10(1.0%)		0.313[0.161-0.605]	0.001
Anxiety	312(8.5%)	220(8.1%)	92(9.5%)		1.176[0.906-1.528]	0.224
Other (specify)	39(1.1%)	27(1.0%)	12(1.2%)		1.250[0.629-2.485]	0.524
COGMEM	3297(89.7%)	2363(87.2%)	934(96.9%)	<0.001	4.572[3.125-6.689]	<0.001
Type of residence						
Single- or multi-family private residence	3446(93.8%)	2556(94.3%)	890(92.3%)		reference	
Retirement community or independent group living	200 (5.4%)	135 (5.0%)	65(6.7%)	0.023	1.383[1.019-1.877]	0.038
Assisted living, adult family home, or boarding home	26(0.7%)	19(0.7%)	7(0.7%)		1.058[0.443-2.525]	0.899
Skilled nursing facility, nursing home, hospital	2(0.1%)	0(0 %)	2(0.2%)		-	0.999
Overall course of decline of cognitive						
Gradually progressive	3313(90.2%)	2405(88.7%)	908(94.2%)		reference	
Stepwise	29(0.8%)	21(0.8%)	8(0.8%)		1.009[0.445-2.286]	0.983
Static	260(7.1%)	221(8.2%)	39(4.0%)	<0.001	0.467[0.330-0.663]	<0.001
Fluctuating	58(1.6%)	50(1.8%)	8(0.8%)		0.424[0.200-0.897]	0.025
Improved	14(0.4%)	13(0.5%)	1(0.1%)		0.204[0.027-1.560]	0.126
NACCMOTF						
No motor symptoms	2962(80.6%)	2165(79.9 %)	797(82.7%)		reference	
Gait disorder	221(6%)	175(6.5%)	46(4.8%)	0.132	0.714[0.511-0.998]	0.049
Falls	89(2.4%)	72(2.7%)	17 (1.8%)		0.641[0.376-1.095]	0.104
Tremor	211(5.7%)	160 (5.9%)	51 (5.3%)		0.866[0.625-1.199]	0.386

Slowness	191 (5.2%)	138 (5.1%)	53 (5.5%)	1.043[0.752-1.447]	0.800
NACCMOTF: Indicate the predominant symptom that was first recognized as a decline in the subject’s motor function. NACCBEHF: Indicate the predominant symptom that was first recognized as a decline in the subject’s behavior. ^a Whites and black or African American individuals made up 95.8% of the subjects. Other races include American Indian or Alaskan Native, Native Hawaiian or Other Pacific Islander, Asian and Other (specify). ^b Dose the companion’s report a decline in the subject’s memory? ^c Non-AD group: the subjects who did not progress to AD at the last follow-up. ^d AD group: the subjects who progressed from MCI to AD.					

eTable.4. The result of hierarchical clustering dendrogram of predictors

Types	The preliminarily selected predictors
1	CRAFTURS, CRAFTDVR, UDSBENTD
2	COGSTAT
3	NACCMOCA, MOCATOTS, MINTTOTS
4	EDUC
5	TRAILA, TRAILB
6	ANIMALS, VEG
7	NACCCOGF, COGMEM
8	CDRGLOB, MEMORY
9	DECIN, DECSUB
10	COGMODE, COURSE
11	SEX, MARISTAT
12	RACE
13	HRATE
14	NACCAGE
15	RESIDENC
16	NACCNE4S
17	NACCBMI

18	BPSYS
19	BEMODE, NACCBEHF
20	NACCGDS, DEP
21	MOMODE, NACCMOTF
22	JUDGMENT, COGJUDG, COMMUN,
23	INDEPEND
24	COGVIS

All the predictors were classified into 24 categories.

eTable.5. AUC values for different years of each neuropsychological measurement scale

	1year	2year	3year	5year	all year	C-index
CDR	0.702	0.662	0.645	0.646	0.651	0.642
Craft Story 21 Recall – Delayed	0.675	0.725	0.718	0.718	0.718	0.695
MOCA	0.715	0.732	0.691	0.678	0.682	0.681
Benson Complex Figure Copy	0.77	0.739	0.728	0.732	0.729	0.713

The models established based on the above four neuropsychological scales all include sociodemographic characteristics, such as age, Body Mass Index (BMI), and years of education.

eTable.6. Overall evaluation criteria for each model-12 predictors (excluding ApoEε4)

	Accuracy	Sensitivity	Specificity	Precision	F1-score	AUC	C-index
Train-Random Survival Forest	0.757	0.794	0.744	0.521	0.629	0.84	0.858
Test-Random Survival Forest	0.682	0.779	0.646	0.449	0.569	0.773	0.779
Train-XGBoost	0.748	0.754	0.746	0.51	0.608	0.816	0.834
Test-XGBoost	0.671	0.785	0.628	0.439	0.563	0.774	0.783
Train-cox	0.697	0.749	0.679	0.45	0.562	0.771	0.788
Test-cox	0.736	0.711	0.745	0.508	0.593	0.799	0.798

Results were calculated based on the regressed predicted probabilities. Cut-offs were determined based on the achievement of the largest Youden index (YI = sensitivity + specificity - 1).

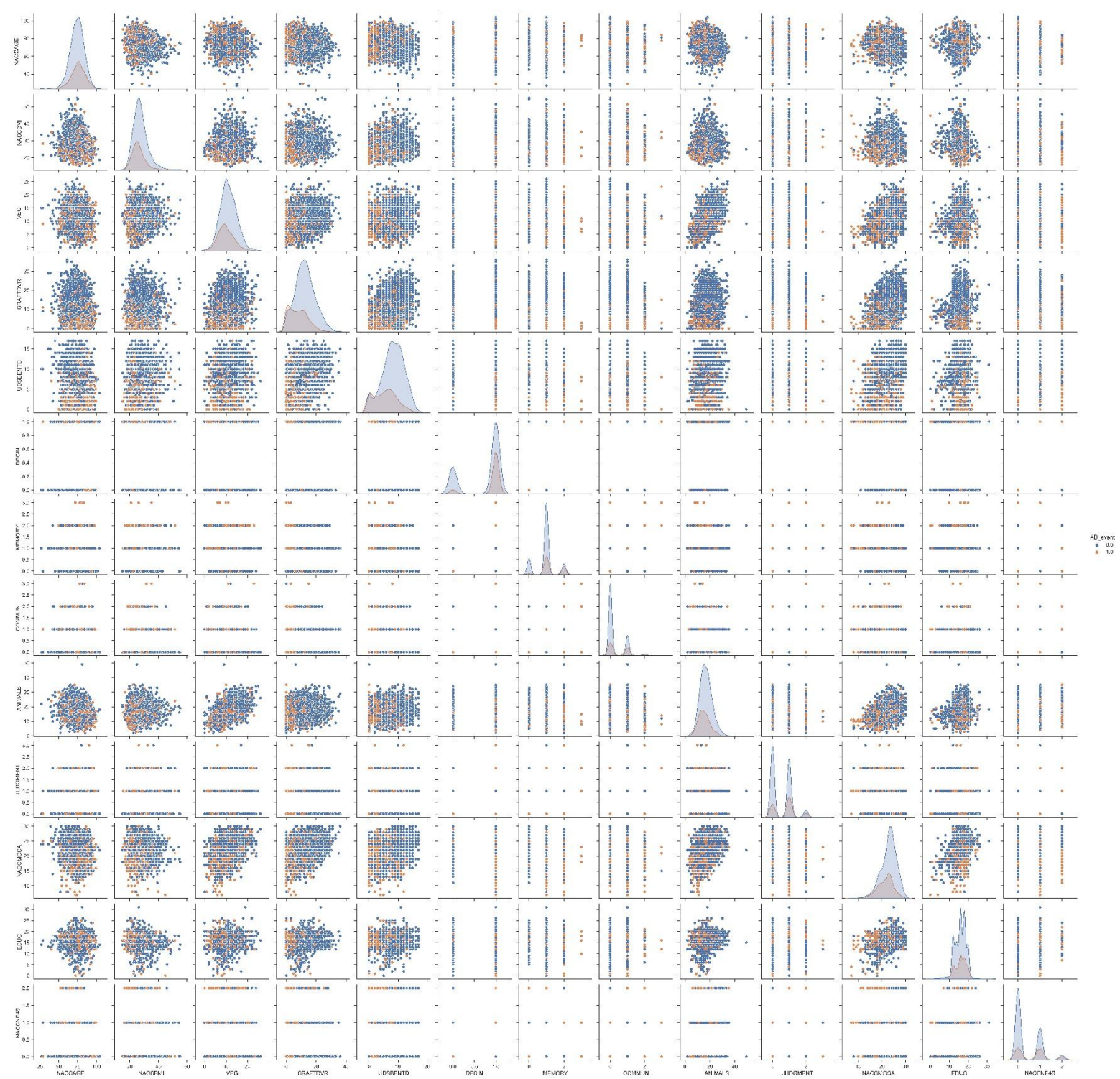
Abbreviations: AD = Alzheimer's Disease, *ApoE ε4*= *Apolipoprotein E (ApoE) ε4*, AUC = Area under the ROC Curve, C-index = Concordance index.

eTable.7. AUC values for different years of each model-12 predictors (excluding ApoEε4)

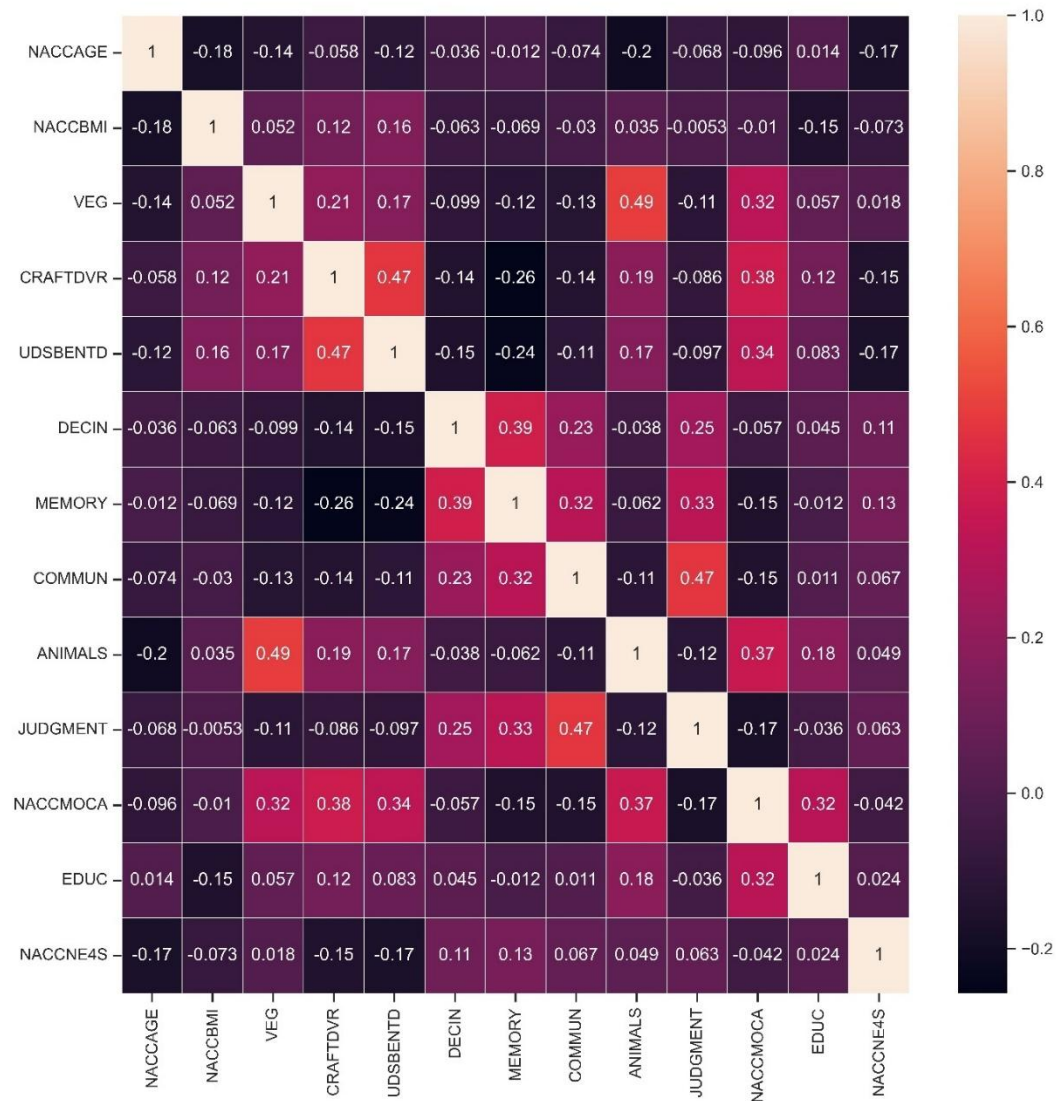
	1year	2year	3year	5year	all year	C-index
Train-Random Survival Forest	0.885	0.879	0.875	0.856	0.84	0.858
Test-Random Survival Forest	0.839	0.823	0.79	0.773	0.773	0.779
Train-XGBoost	0.873	0.854	0.848	0.829	0.816	0.834
Test-XGBoost	0.861	0.826	0.794	0.774	0.774	0.783
Train-cox	0.826	0.811	0.801	0.784	0.771	0.788
Test-cox	0.858	0.841	0.815	0.798	0.799	0.798

Results were calculated based on the regressed predicted probabilities. Cut-offs were determined based on the achievement of the largest Youden index (YI = sensitivity + specificity - 1).

Abbreviations: AD = Alzheimer's Disease, *ApoE ε4*= *Apolipoprotein E (ApoE) ε4*, AUC = Area under the ROC Curve, C-index = Concordance index.



eFig. 1. Pairwise relationship diagrams of each predictor in the dataset



eFig. 2. Correlations heatmap of 13 predictors

The 13 predictors in models were as follows: 3 sociodemographic characteristics (age, BMI, and years of education); the neuropsychological measurement scales, including the Montreal Cognitive Assessment (MoCA) score and the scores of the memory, judgment, and social activity dimensions in the Clinical Dementia Rating (CDR); the Craft Story 21 Recall–Delayed (CRAFTDVR), Benson Complex Figure Copy (UDSBENTD) and Category Fluency (VEG and ANIMALS) scores; companion reports on whether the subjects’ memory ability decreased compared to before (DECIN); and the number of ApoEε4 alleles.